Employee Attrition Prediction - Project Report

1. Project Overview

This project builds an AI-powered system to predict whether an employee is likely to leave the company (attrition) using the IBM HR Analytics Employee Attrition & Performance dataset. The system covers data preprocessing, model building, evaluation, and insights visualization.

2. Dataset

Source: IBM HR Analytics Employee Attrition Dataset (Kaggle). It contains employee demographic, job role, and performance data with the target variable 'Attrition' (Yes/No).

3. Steps Performed

- Data Understanding & Preprocessing: Missing value handling, categorical encoding, outlier detection, scaling.
- Exploratory Data Analysis (EDA): Created visualizations showing attrition patterns by age, job role, and income.
- Model Building: Implemented Logistic Regression and Random Forest models with hyperparameter tuning.
- Evaluation: Used Accuracy, Precision, Recall, F1 Score, Confusion Matrix, and ROC-AUC metrics.
- Insights: Found key factors affecting attrition such as Overtime, Job Role, Monthly Income, and Job Satisfaction.

4. Research Brief: Agentic AI (Autonomous AI Agents)

- What is it? AI systems capable of independent decision-making, planning, and action execution to meet goals.
- Why it matters? Enables scalable automation, reduces human intervention, and improves operational efficiency.
- Use case: AI agents for automated customer service handling queries from start to resolution.
- Challenges: Reliability, ethical concerns, and ensuring alignment with human intentions.

5. Conclusion

The project successfully demonstrated the process of building and evaluating a machine learning model for employee attrition prediction. Random Forest achieved the best performance, and the research brief explored the relevance of Agentic AI in modern business contexts.